

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

Never turn the engine in the direction opposite to that of normal operation

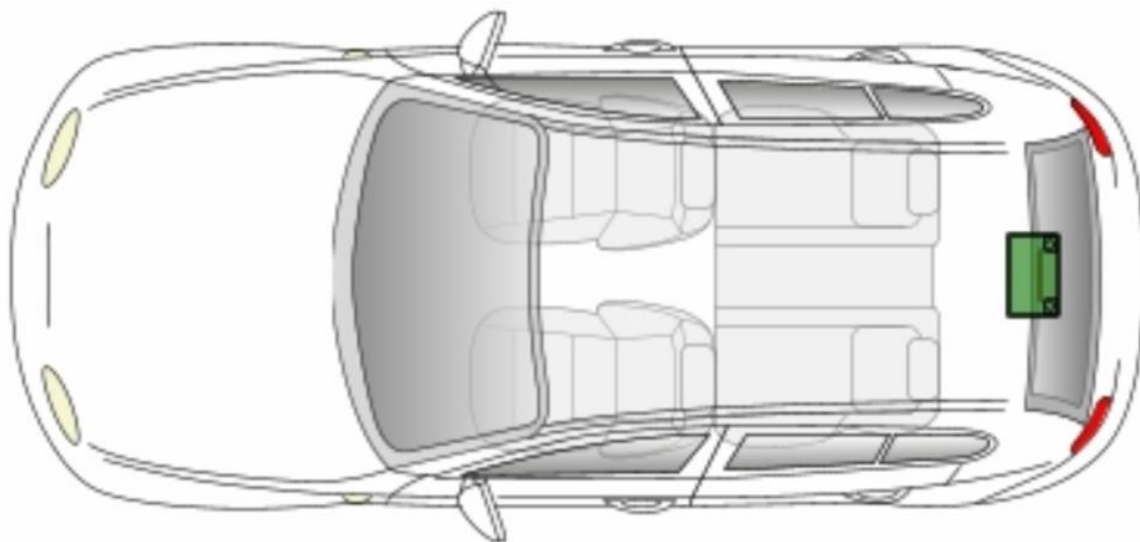
Note: Avoid bending, torsion or traction

Always use new gaskets

General

Before disconnecting the battery cable, check the audio system security code

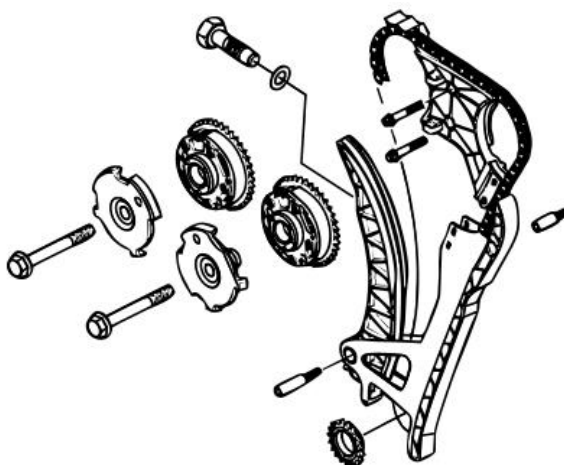
Battery location



Luggage compartment

Timing chain layout

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Removal

Disconnect the battery earth cable

Remove the fan cowl

Remove the engine upper cover

Remove the ignition coils

Disconnect the injection wiring

Remove the injectors

Remove the strut brace connecting the shock absorbers

Remove the air duct

Remove the cylinder head cover

Remove the spark plugs

Remove the chain tensioner piston

Compress the chain tensioner piston twice to drain the oil

Remove the engine lower cover

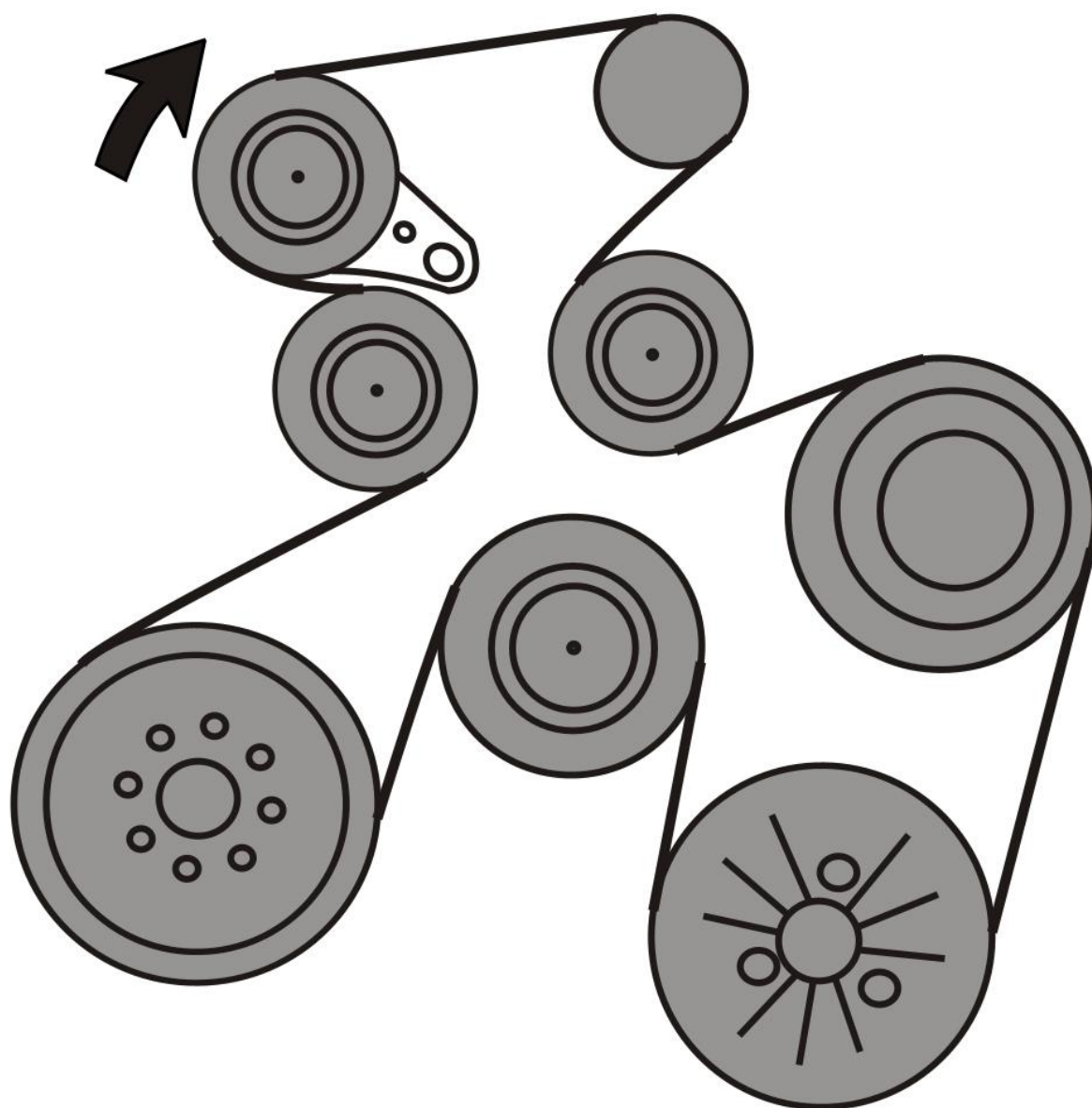
Turn the tensioner pulley clockwise to release the tension

Lock the tensioner; use a locking pin (11 3 340)

Remove the ancillary drive belt

Drive belt layout (With air conditioning)

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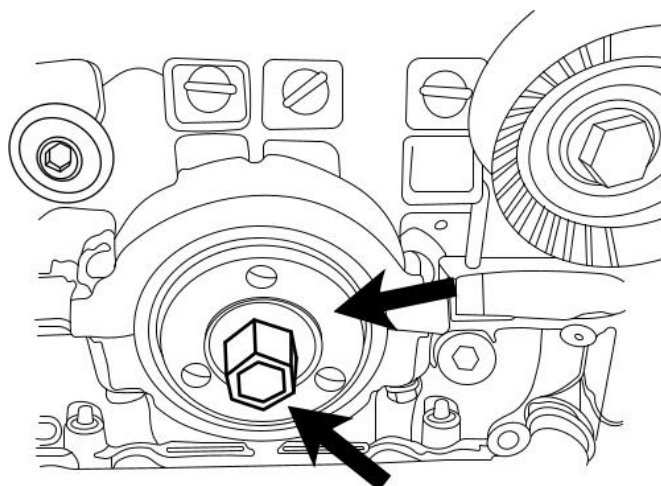


Note: At removal, mark the direction of rotation; when refitting ensure the drive belt rotates in the same direction

Remove the crankshaft pulley

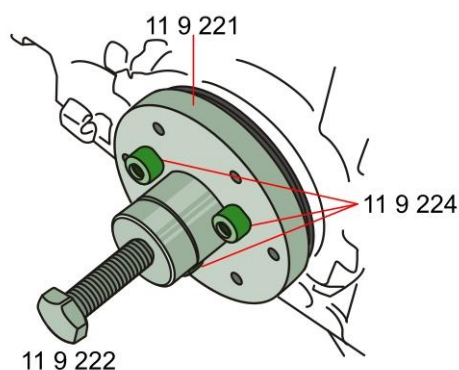
Note: Do not remove

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Fit the special tools (11 9 221, 11 9 222, 11 9 224)

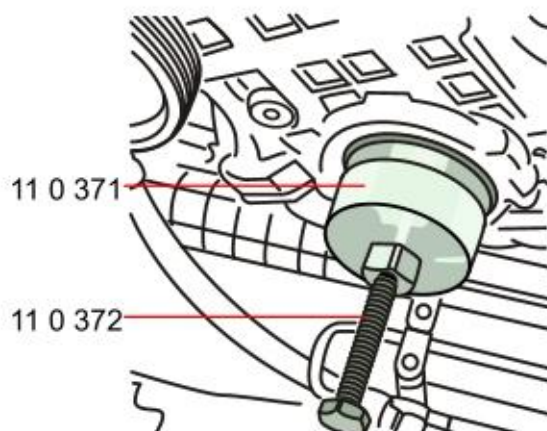
Tighten the special tool (11 9 224 - 20 Nm)



Remove the special tools (11 9 221, 11 9 222, 11 9 224)

Fit the special tools (11 0 371, 11 0 372)

Tighten the special tool (11 0 371 - 80 Nm)



Remove and discard the crankshaft oil seal

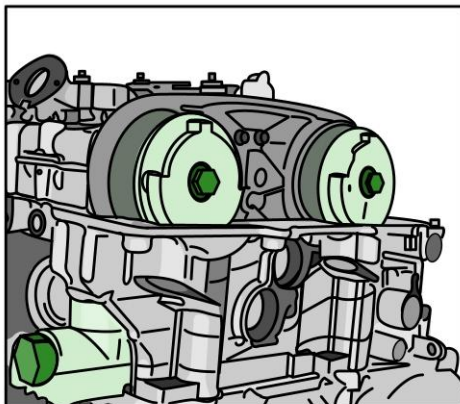
Remove the locking pin

Remove the ancillary drive belt tensioner (11 3 340)

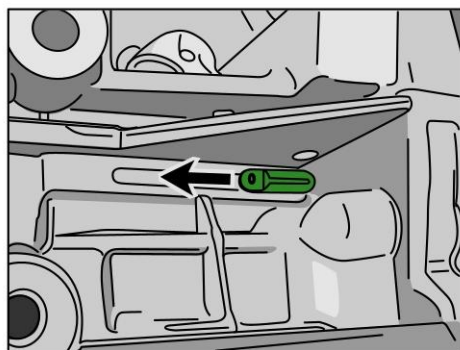
Remove the inlet camshaft timing adjuster

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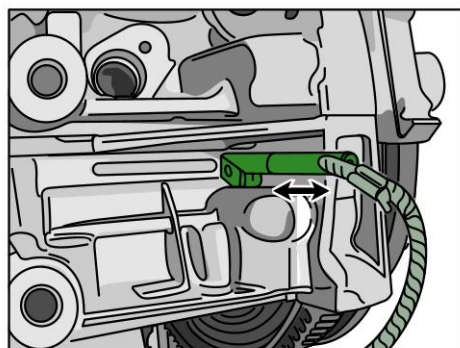
Remove the exhaust camshaft timing adjuster



Remove the access plug

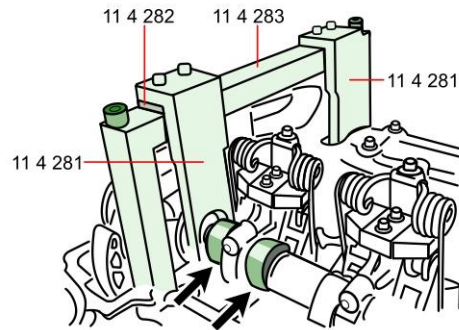


Fit the crankshaft TDC locking pin (11 0 300)



Fit the camshaft locking tools (11 4 281, 11 4 282, 11 4 283)

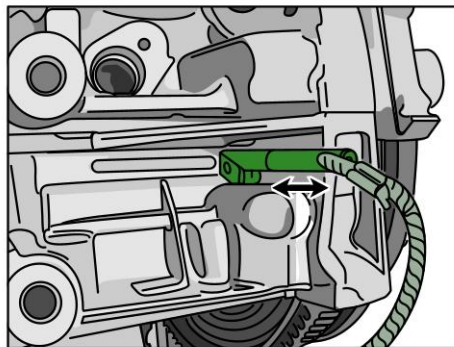
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Lock the flywheel

Use the special tools (11 8 660, 11 8 160)

Remove the crankshaft locking pin



Remove the crankshaft gearwheel centre bolt

Remove the crankshaft hub

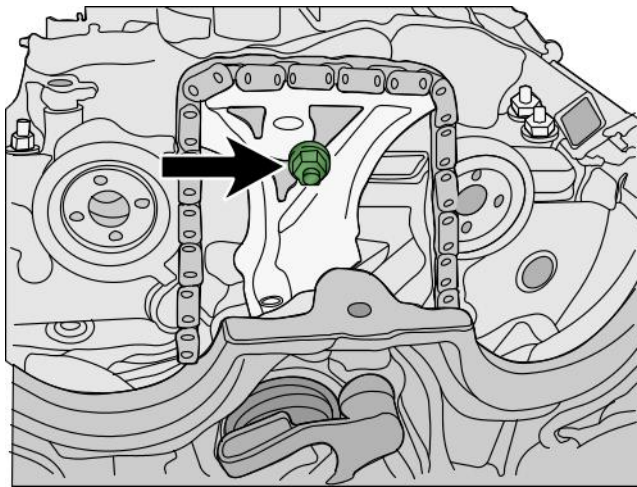
Bearing pins for guide and tensioner rails:

Remove the access plugs

Remove the guide pins

Remove the timing chain guide bolts from the cylinder head

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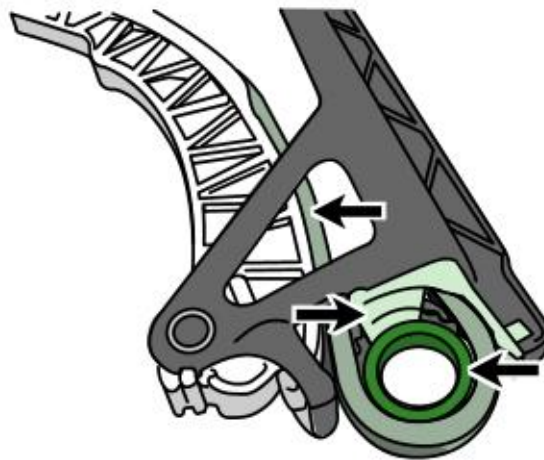
Remove the chain guide with the timing chain, guide rail and tensioner rail

Installation

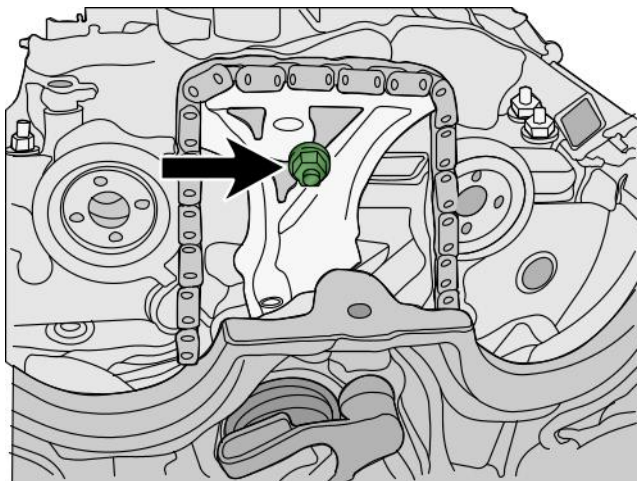
Crankshaft gearwheel:

Fit the shims

Refit the chain guide with the timing chain, guide rail and tensioner rail



Refit the timing chain guide bolts to the cylinder head (M6 x 35: 9 Nm)



Refit the crankshaft hub

Fit the special tool (00 9 140)

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Tighten the crankshaft gearwheel bolt (100 Nm + 360°)

Use new bolt(s)

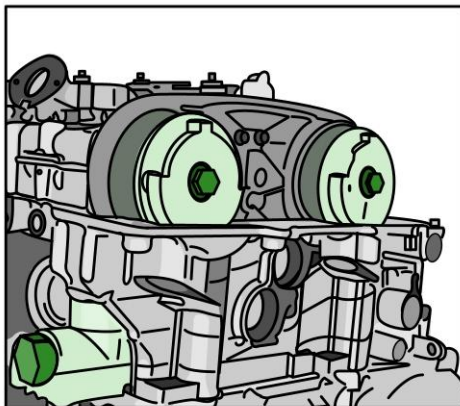
Bearing pins for guide and tensioner rails:

Refit the guide pins (M7: 14 Nm, M8: 20 Nm)

Refit the access plugs (M18 x 1.5: 25 Nm)

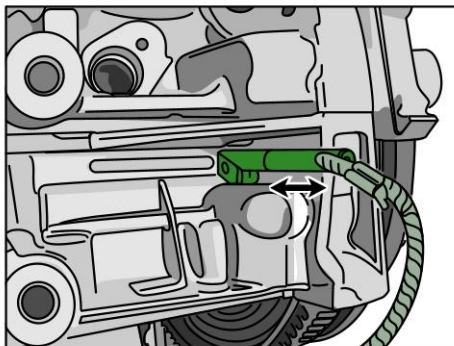
Refit the exhaust camshaft adjuster (AUS/EX)

Refit the inlet camshaft timing adjuster (EIN/IN)



Tighten the camshaft gearwheel bolts lightly until there is no play

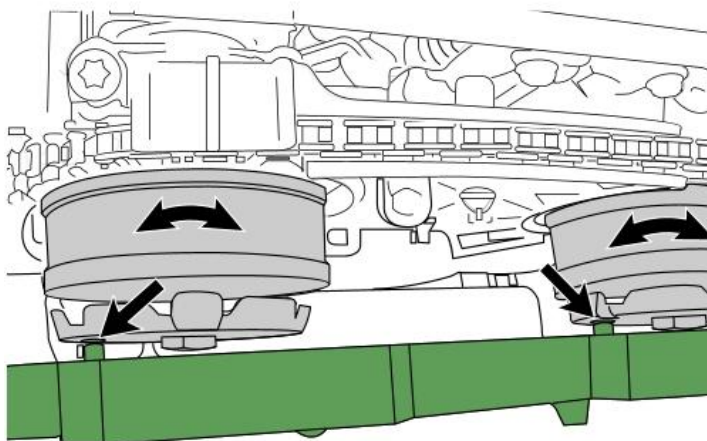
Fit the crankshaft TDC locking pin (11 0 300)



Fit the camshaft alignment tool (11 8 520)

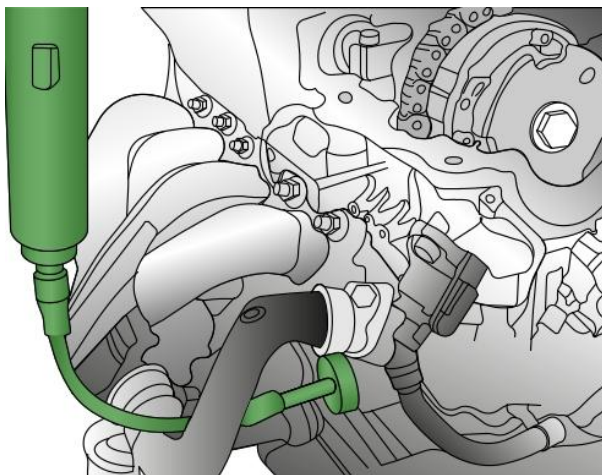
Align the camshaft gearwheel marks as shown

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Pre-tension the timing chain (0.6 Nm)

Use the special tools (00 9 250, 11 9 340)

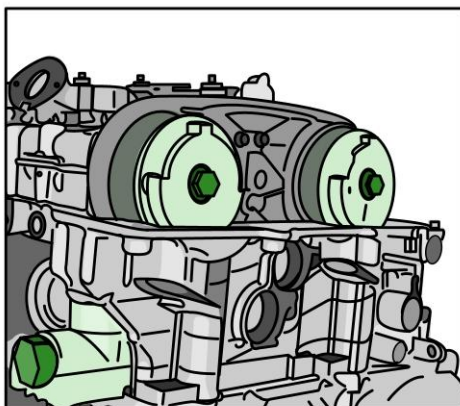


Tighten the camshaft gearwheel bolts (20 Nm + 180°)

Use new bolt(s)

Refit the chain tensioner piston (55 Nm)

Note: Fit a sealing ring



Remove all the locking tools

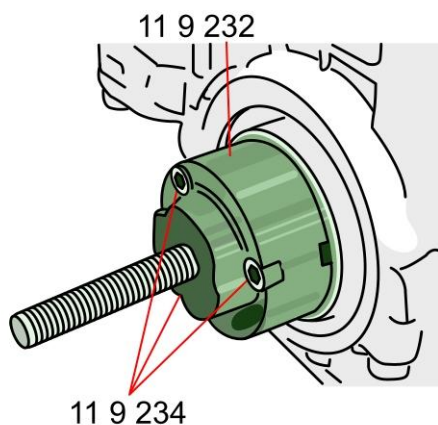
Turn the engine 2 rotations by hand

Check the timing again

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Refit the ancillary drive belt tensioner (M10 x 110: 40 Nm)

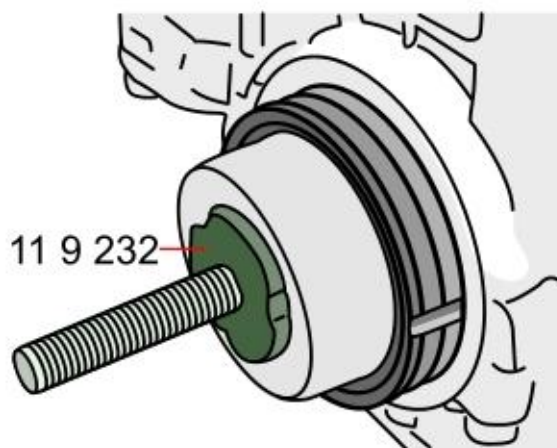
Fit the special tools (11 9 232, 11 9 234)



Apply grease

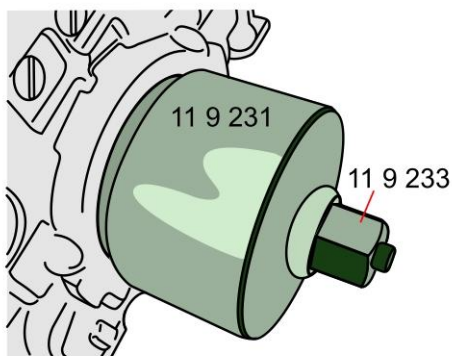
Fit the crankshaft seal over the special tool

Note: The seal should be aligned with the timing cover



Refit the crankshaft front radial seal

Use the special tool (11 9 231, 11 9 233)



Apply sealant

Refit the crankshaft pulley (25 Nm)

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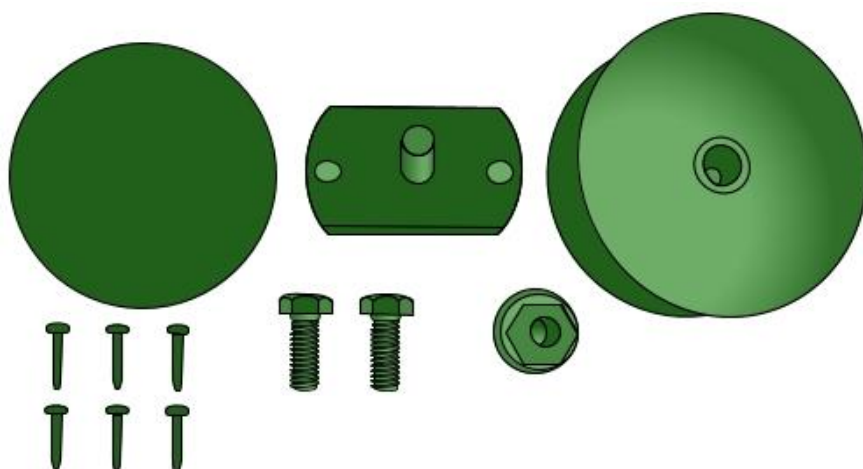
Turn the tensioner clockwise
Lock the tensioner; use a locking pin (11 3 340)
Fit the ancillary drive belt
Remove the tensioner locking pin (11 3 340)
Refit the engine lower cover
Refit the spark plugs (M12 x 1.25: 23 Nm, M14 x 1.25: 30 Nm)
Refit the cylinder head cover (9 Nm)
 Use new bolt(s)
Refit the air duct
Refit the strut brace connecting the shock absorbers
Refit the injectors
Reconnect the injection wiring
Refit and connect the ignition coils
Refit the engine upper cover
Refit the fan cowling
Reconnect the battery earth cable

Torque settings

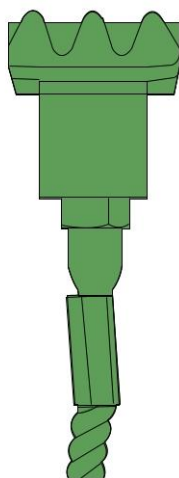
Camshaft gearwheel
 Camshaft gearwheel
 Renew the bolts
 Stage 1 20 (Nm)
 Stage 2 180 (°)
Crankshaft pulley
 Crankshaft pulley (Outer bolts) 35 (Nm)
Crankshaft gearwheel
 Crankshaft gearwheel (Centre bolt)
 Renew the bolts (Threads lightly oiled)
 Stage 1 100 (Nm)
 Stage 2 360 (°)
Spark plugs
 Spark plugs (M12) 20 - 26 (Nm)
 Spark plugs (M14) 27 - 33 (Nm)
Injector
 Injector 11 - 15 (Nm)
Fuel rail
 Fuel rail 11 - 15 (Nm)
Valve cover
 Valve cover 8.5 (Nm)
Ancillary drive belt tensioner pulley
Timing chain tensioner
 Timing chain tensioner (Upper)
 Renew the seal
 Stage 1 55 (Nm)
Timing chain guide
 Timing chain guide (Upper) 14 (Nm)
 Timing chain guide (Lower) 20 (Nm)
Chain guide: (M6 x 35: 9 Nm)
Access plug: (M18 x 1.5: 25 Nm)

Special tools

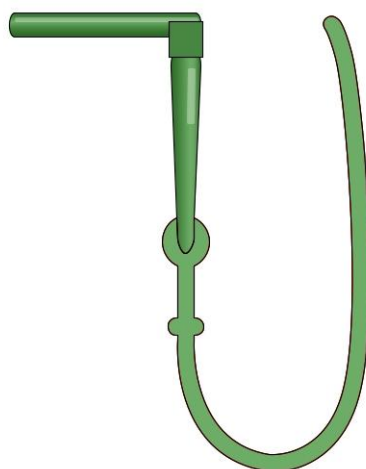
Extractor: (11 9 221, 11 9 222, 11 9 224, 11 0 371, 11 0 372)
Crankshaft front radial seal fitting tool: (11 9 231, 11 9 232, 11 9 233, 11 9 234)

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Flywheel locking tool: (11 8 660, 11 8 160)

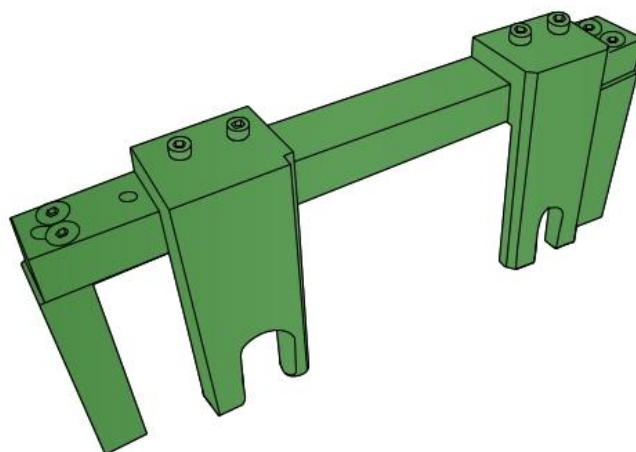


Crankshaft locking tool: (11 0 300)

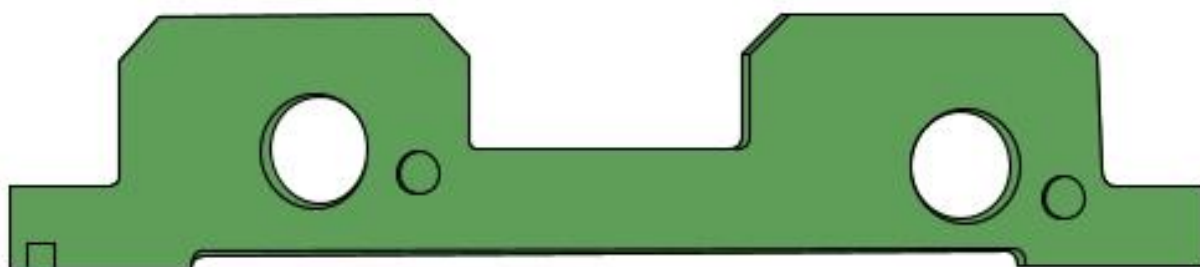


Camshaft locking tool: (11 4 281, 11 4 282, 11 4 283)

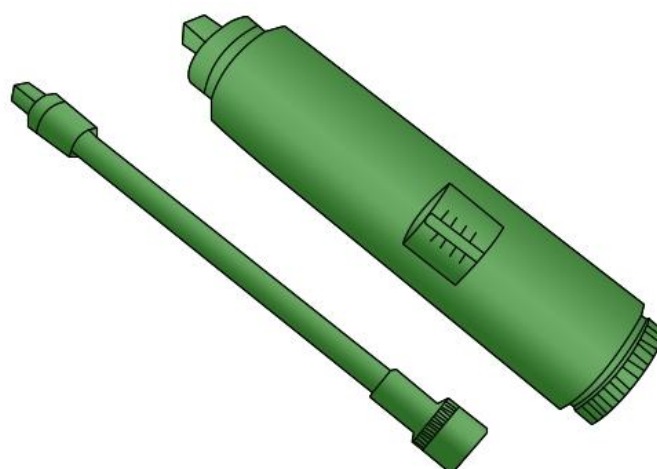
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Camshaft alignment tool: (11 8 520)

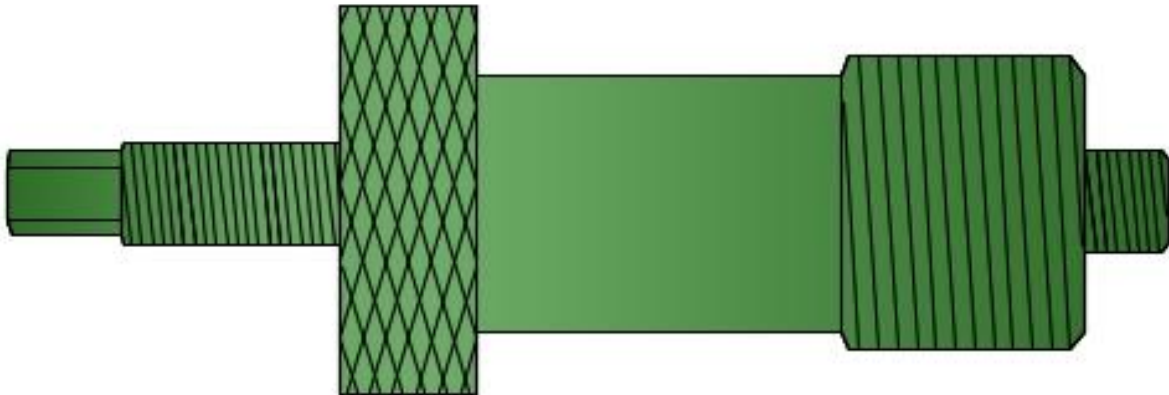


Torque wrench: (00 9 250)



Adaptor: (11 9 340)

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Tensioner locking pin: (11 3 340)

